Kinematics

Def A lightcone basis is a basis $\{\bar{n}, n\}$ that satisfies

$$n^2 = \bar{n}^2 = 0 \text{ and } n \cdot \bar{n} = 1$$

The transverse projector $g_T^{\mu\nu} = g^{\mu\nu} - \bar{n}^\mu n^\nu - \bar{n}^\nu n^\mu$

Def Let $\{n, \bar{n}\}$ be a lightcone basis. Then a 4-vector P can be decomposed as $p^\mu = p^+ \bar{n}^\mu + p^- n^\mu + p_T{}^\mu$. $\{n, \bar{n}\}$ is a P-collinear basis if

$$[p^+, p^-, \vec{p}_T] \sim Q[1, \lambda^2, \lambda]$$

, where Q is a hard scale and $\lambda = \frac{M}{Q}$

Def For a 4-vector P, a P-collinear basis $\{\bar{n}, n\}$ is a P-intrinsic basis if

$$g_T^{\mu\nu} P_\nu = 0$$

i.e. transverse component $P_T^\mu = 0$; here $g_T^{\mu\nu} \equiv g^{\mu\nu} - \bar{n}^\mu n^\nu - \bar{n}^\nu n^\mu$

Prop Given a 4-momentum P, its P-collinear basis is not unique. For any P-collinear basis $\{\bar{n}, n\}$, we can apply the following Lorentz transformations to obtain a new P-collinear basis $\{\bar{n}', n'\}$.

Longitudinal rescaling	transverse rotation	tilt $\bar{n}$	tilt n
$\bar{n}' = e^\alpha$ $n = e^{-\alpha}$	$e_i' \to R_{ij} e_j$	$\bar{n}' = \bar{n} + b_T - \frac{b_T^2}{2} n$	$n = n + b_T - \frac{b_T^2}{2} \bar{n}$
$\lvert\alpha\rvert \lesssim 1$		$g_T^{\mu\nu} b_{T\nu} = 0$, $\lvert b_T \rvert \sim \lambda$	

Prop Consider we tilt $\bar{n}/n$ by b_T to obtain $\bar{n}'$. The the corresponding transformations are

	tilt $\bar{n}$	tilt n
$\bar{n}'$	$\bar{n}' = \bar{n} + b_T - \frac{b_T^2}{2} n$	$\bar{n}' = \bar{n}$
n'	$n' = n$	$n' = n + b_T - \frac{b_T^2}{2} \bar{n}$
e_i'	$e_i' = e_i - (e_i \cdot b_T) n$	$e_i' = e_i - (e_i \cdot b_T) \bar{n}$

$a^{+\prime}$	$a^{+\prime} = a^+$	$a^{+\prime} = a^+ + a_T\cdot b_T - \frac{b_T^2}{2} a^-$
$a^{-\prime}$	$a^{-\prime} = a^- + a_T\cdot b_T - \frac{b_T^2}{2} a^+$	$a^{-\prime} = a^-$
$a_T^{\mu\prime}$	$a_T^{\mu\prime} = a_T^\mu - a^+ b_T^\mu - b_T\cdot(a_T - a^+ b_T)\, n^\mu$	$a_T^{\mu\prime} = a_T^\mu - a^- b_T^\mu - b_T\cdot(a_T - a^- b_T)\, \bar{n}^\mu$
$g_T^{\prime\mu\nu}$	$g_T^{\prime\mu\nu} = g_T^{\mu\nu} - b_T^\mu n^\nu - b_T^\nu n^\mu + b_T^2 n^\mu n^\nu$	$g_T^{\prime\mu\nu} = g_T^{\mu\nu} - b_T^\mu \bar{n}^\nu - b_T^\nu \bar{n}^\mu + b_T^2 \bar{n}^\mu \bar{n}^\nu$
$\varepsilon^{\prime\mu\nu}$	$\varepsilon^{\prime\mu\nu} = \varepsilon^{\mu\nu} + \varepsilon^{\mu\nu\rho\sigma} b_{T\rho} n_\sigma$	$\varepsilon^{\prime\mu\nu} = \varepsilon^{\mu\nu} - \varepsilon^{\mu\nu\rho\sigma} b_{T\rho} \bar{n}_\sigma$

Quark TMDPDF (large P^+)

	P-Intrinsic	Global
$\bar n'$		$\bar n_P = \bar n + \frac{P_T}{P^+} - \frac{P_T^2}{2P^{+2}} n$
e_i'		$e_{Pi} = e_i - \frac{e_i \cdot P_T}{P^+} n$
P	$P = P^+ \bar n_P + P_P^- n$	$P = P^+ \bar n + P^- n + P_T$
S	$S = S_L \frac{P^+}{M} \bar n_P + S_P^- n + S_\perp$	$S = S_L \frac{P^+}{M} \bar n + S^- n + [(S_L \frac{P_T}{M} + S_\perp) + \frac{P_T \cdot S_\perp}{P^+} n]$
$S_\perp$		$S_\perp = S_T - S_L \frac{P_T}{M} - \frac{1}{P^+}(P_T \cdot S_T - S_L \frac{P_T^2}{M}) n$
k	$k = x P^+ \bar n_P + k_P^- n + k_\perp$	$k = x P^+ \bar n + k^- n + [(x P_T + k_\perp) + \frac{P_T \cdot k_\perp}{P^+} n]$
$k_\perp$		$k_\perp = k_T - x P_T - \frac{1}{P^+}(P_T \cdot k_T - x P_T^2) n$
$g_T'^{\mu\nu}$	$g_T^{\mu\nu} = g^{\mu\nu} - \bar n_P^\mu n^\nu - \bar n_P^\nu n^\mu$	$g_{TP}^{\mu\nu} = g_T^{\mu\nu} - \frac{P_T^\mu n^\nu + n^\mu P_T^\nu}{P^+} + \frac{P_T^2}{P^{+2}} n^\mu n^\nu$
$\epsilon'^{\mu\nu}$	$\epsilon_P^{\mu\nu} = \epsilon^{\mu\nu\rho\sigma} \bar n_{P\rho} n_\sigma$	$\epsilon_P^{\mu\nu} = \epsilon^{\mu\nu} + \epsilon^{\mu\nu\rho\sigma} \frac{P_{T\rho} n_\sigma}{P^+} = \epsilon^{\mu\nu} + \frac{1}{P^+}(\tilde P_T^\mu n^\nu - \tilde P_T^\nu n^\mu)$

Definition

Given a global P-collinear basis $\{\bar{n}, n\}$, sps

$$P^\mu = [P^+, P^-, P_T^\mu]$$

$$S^\mu = \left[S_L \frac{P^+}{M}, S^-, S_L \frac{P_T^\mu}{M} + S_\perp^\mu\right]$$

$$k^\mu = [xP^+, k^-, xP_T^\mu + k_\perp^\mu]$$

, where $k_\perp, S_\perp$ are the intrinsic transverse momentum/spin. We define TMDPDF in the hadron intrinsic frame $\{\bar{n}_P, n\}$, where $\bar{n}_P^\mu = \bar{n}^\mu - \frac{P_T^2}{2P^{+2}} n^\mu + \frac{P_T^\mu}{P^+}$, and

$$P^\mu = P^+ \bar{n}_P^\mu + P_P^- n^\mu$$

$$S^\mu = S_L \frac{P^+}{M} \bar{n}_P^\mu + S_P^- n^\mu + S_\perp^\mu$$

$$k^\mu = xP^+ \bar{n}_P^\mu + k_P^- n^\mu + k_\perp^\mu$$

Def $\Phi_{ab}(x, k_\perp; P, S)$

$$= \int \frac{d\xi^- d^2\xi_T}{(2\pi)^3} e^{i(xP^+\xi^- - \vec{k}_\perp \cdot \vec{\xi}_\perp)} \langle P, S | \bar{\psi}_b(0) W(0,\xi) \psi_a(\xi) | P, S \rangle \Big|_{\xi^+=0}$$

$$= \frac{1}{2}\left\{ f_1 \bar{\not{n}}_P + f_{1T}^\perp \frac{\varepsilon_{\mu\nu\rho\sigma}\gamma^\mu \bar{n}_P^\nu k_\perp^\rho S_\perp^\sigma}{M} + g_{1S}\gamma_5 \bar{\not{n}}_P + h_{1T}\, i\sigma^{\bar{n}_P S_\perp}\gamma^5 + \frac{h_{1S}^\perp}{M} i\sigma^{\bar{n}_P k_\perp}\gamma^5 + \frac{h_1^\perp}{M}\sigma^{k_\perp \bar{n}_P} \right\}$$

$$= \frac{1}{2}\left\{ \left[f_1 - f_{1T}^\perp \frac{\hat{k}_\perp \cdot S_\perp}{M}\right] \bar{\not{n}}_P + g_{1S}\gamma^5 \bar{\not{n}}_P + \left[h_{1T} S_{\perp\mu} + h_{1S}^\perp \frac{k_{\perp\mu}}{M} + h_1^\perp \frac{\tilde{k}_{\perp\mu}}{M}\right] i\sigma^{\bar{n}_P \mu}\gamma^5 \right.$$

, where

$$g_{1S} = S_L g_{1L} - \frac{k_\perp \cdot S_\perp}{M} g_{1T}, \quad h_{1S}^\perp = S_L h_{1L}^\perp - \frac{k_\perp \cdot S_\perp}{M} h_{1T}^\perp$$

Prop Let $\Phi^\Gamma(z, k_\perp) = \frac{1}{2} \mathrm{Tr}[\Gamma \Phi(z, k_\perp)]$. Then,

$\Phi^{\not{n}}$	$\Phi^{\not{n}\gamma^5}$	$\Phi^{i\sigma^{nk}\gamma^5}$	$\Phi^{i\sigma^{n\tilde{k}}\gamma^5}$	$\Phi^{i\sigma^{nS}\gamma^5}$
$f_1 - f_{1T}^\perp \frac{\hat{k}_\perp \cdot S_\perp}{M}$	g_{1S}	$-(h_{1T} S_\perp \cdot k_\perp + h_{1S}^\perp \frac{k_\perp^2}{M})$	$-(h_{1T} S_\perp \cdot \tilde{k}_\perp + h_1^\perp \frac{\tilde{k}_\perp^2}{M})$	$-h_{1T} S_\perp^2 - \frac{1}{M}(h_{1S}^\perp k_\perp \cdot S_\perp + h_1^\perp \hat{k}_\perp \cdot S_\perp)$

Normalization

Prop At LO partonic level, $f_1 = g_{1L} = h_{1T} = \delta(1-x)\,\delta^2(\vec{k}_\perp)$, while others are 0.

proof Recall $\begin{cases} \psi_i(y)|q(P,S)\rangle = u_i(P,S)\,e^{-iP\cdot y}|0\rangle \\ \langle q(P,S)|\bar\psi_j(y) = \bar u_j(P,S)\,e^{iP\cdot y}\langle 0| \end{cases}$

$\Phi_{ab}(x, k_\perp; P, S)$

$= \int \frac{d\xi^- d^2\xi_T}{(2\pi)^3} e^{ik\cdot\xi} \langle P,S|\bar\psi_b(0)\psi_a(\xi)|P,S\rangle \Big|_{\xi^+=0,\ k^+=xP^+}$

$= \int \frac{d\xi^- d^2\xi_T}{(2\pi)^3} e^{ik\cdot\xi} [u_S(P)\bar u_S(P)]_{ab}\, e^{-iP\cdot\xi}$

$= [u_S(k)\bar u_S(k)]_{ab}\, \delta(k^+ - P^+)\, \delta^2(\vec{k}_\perp - \vec{P}_\perp)$

$= \frac{1}{2P^+}(\not{P} + m)(1 + \gamma^5 \not{S})\, \delta(1-x)\, \delta^2(\vec{P}_\perp - \vec{k}_\perp)$. Recall $\vec{P}_\perp = 0$.

① $\not{P} = P^+ \bar{\not{n}}_P + P^- \not{n}$. At twist 2, $\not{P} = P^+ \bar{\not{n}}_P$

$\frac{1}{2}\mathrm{Tr}[\not{n}\,\Phi] = \frac{1}{4P^+} P^+ \mathrm{Tr}[\not{n}\,\bar{\not{n}}_P]\,\delta(1-x)\,\delta^2(\vec{k}_\perp) = \delta(1-x)\,\delta^2(\vec{k}_\perp)$

② $m\gamma^5\not{S} = m\gamma^5(S^+\bar{\not{n}}_P + S^-\not{n} - S_\perp^i\gamma^i)$. At twist 2, $m\gamma^5\not{S} = S_L P^+ \gamma^5 \bar{\not{n}}$

$\frac{1}{2}\mathrm{Tr}[\not{n}\gamma^5\Phi] = \frac{1}{4P^+} S_L P^+ \mathrm{Tr}[\not{n}\gamma^5\gamma^5\bar{\not{n}}]\,\delta(1-x)\,\delta^2(\vec{k}_\perp) = S_L\,\delta(1-x)\,\delta^2(\vec{k}_\perp)$

③ $\not{P}\gamma^5\not{S} = -\not{P}\not{S}\gamma^5 = -P_\mu S_\nu \gamma^\mu\gamma^\nu\gamma^5$. At twist 2,

$\not{P}\gamma^5\not{S} = -P_- S_i \bar{\not{n}}_P \gamma^i\gamma^5 = -P^+ S^i \gamma^i \bar{\not{n}}_P \gamma^5 = P^+ S_\perp^i\, i\sigma^{i-}\gamma^5$

$\frac{1}{2}\mathrm{Tr}[i\sigma^{i+}\gamma^5\Phi] = \frac{1}{4P^+} P^+ S_\perp^i\, \mathrm{Tr}[i\sigma^{i+}\gamma^5\, i\sigma^{i-}\gamma^5]\,\delta(1-x)\,\delta^2(\vec{k}_\perp)$

$= S_\perp^i\,\delta(1-x)\,\delta^2(\vec{k}_\perp)$ □

Cut - Vertex

$$\Phi^{\Gamma}$$

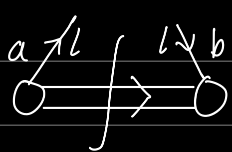
$$= \int \frac{d\xi^- d^2\xi_T}{2(2\pi)^3} e^{i(xP^+\xi^- - k_\perp \cdot \xi_\perp)} \mathrm{Tr}\left[\Gamma \langle P,S|\bar{\psi}_b(0) W(0,\xi)\psi_a(\xi)|P,S\rangle|_{\xi^+=0}\right]$$

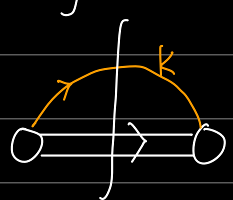
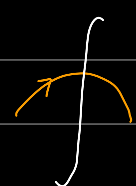
$$= \int dl^+ d^2 l_\perp\, \delta(xP^+ - l^+)\, \delta^2(\vec{l}_\perp - \vec{k}_\perp)$$

$$\int \frac{d\xi^- d^2\xi_T}{2(2\pi)^3} e^{i(xP^+\xi^- - \vec{k}_\perp \cdot \vec{\xi}_\perp)} \int d\xi^+ \int \frac{dl^-}{2\pi} e^{il^-\xi^+} \mathrm{Tr}\left[\Gamma \langle P,S|\bar{\psi}_b(0) W(0,\xi)\psi_a(\xi)|P,S\rangle\right]$$

$$= \int \frac{d^4 l}{(2\pi)^4} \mathrm{Tr}\Big[\underbrace{\frac{1}{2P^+}\delta\left(\frac{l^+}{P^+} - x\right)\delta^2(\vec{l}_\perp - \vec{k}_\perp)\Gamma}_{\text{cut vertex}} \underbrace{\int d^4\xi\, e^{il\cdot\xi} \langle P,S|\bar{\psi}_b(0) W(0,\xi)\psi_a(\xi)|P,S\rangle}_{\hat{T}_{ab}(l;P,S)} \Big]$$

Diagramatically,

$\hat{T}_{ab}(l;P,S) =$ 

$\Phi^{\Gamma}(x, k_\perp; P; S) =$  , where  $= \frac{1}{2P^+}\delta\left(\frac{l^+}{P^+} - x\right)\delta^2(\vec{l}_\perp - \vec{k}_\perp)\Gamma$

$$\Phi(x, \vec{k}_\perp; P, S) = \int \frac{d^4 l}{(2\pi)^4} \frac{1}{P^+}\delta\left(\frac{l^+}{P^+} - x\right)\delta^2(\vec{l}_\perp - \vec{k}_\perp) T_\Phi(l;P,S)$$

Quark Collinear PDF (large P^+)

Def $\Phi_{ab}(x;P,S)$

$$= \int \frac{d\xi^-}{2\pi} e^{ixP^+\xi^-} \langle P,S|\bar{\psi}_b(0)W(0,\xi^-)\psi_a(\xi^-)|P,S\rangle$$

$$= \frac{1}{2}\left[f_1 \bar{\not{n}}_P + S_L g_{1L}\gamma^5 \bar{\not{n}}_P + h_1 i\sigma^{\bar{n}_P S_\perp}\gamma^5\right]$$

Prop Let $\Phi^\Gamma(x) = \frac{1}{2}\mathrm{Tr}[\Gamma\Phi(x)]$. Then,

$\Phi^{\not{n}}$	$\Phi^{\not{n}\gamma^5}$	$\Phi^{i\sigma^{nS}\gamma^5}$
f_1	$S_L g_{1L}$	$-H_1 S_\perp^2$

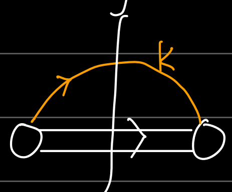
Cut-Vertex

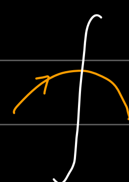
Φ_{ij}

$$= \int \frac{d^4 l}{(2\pi)^4} \underbrace{\delta(l^+ - k^+)}_{\text{cut vertex}} \underbrace{\int d^4\xi\, e^{il\cdot\xi} \langle P,S|\bar{\psi}_b(0)W(0,\xi)\psi_a(\xi)|P,S\rangle}_{\hat{T}(l;P,S)}$$

Diagramatically,

$\hat{T}_{ab}(l;P,S) =$ 

$\Phi^\Gamma(x;P;S) =$ 

, where  $= \frac{1}{2}\delta(l^+ - k^+)\Gamma$

$$\Phi(x;P,S) = \int \frac{d^4 l}{(2\pi)^4} \frac{1}{P^+}\delta\left(\frac{l^+}{P^+} - x\right) T_\Phi(l;P,S)$$

Quark TMDPDF (large P^-)

	P-Intrinsic	Global
n'		$n_P = n + \frac{P_T}{P^-} - \frac{P_T^2}{2P^{-2}}\bar{n}$
e_i'		$e_{Pi} = e_i - \frac{e_i \cdot P_T}{P^-}\bar{n}$
P	$P = P^- n_P + P_P^+ \bar{n}$	$P = P^- n + P^+ \bar{n} + P_T$
S	$S = S_L \frac{P^-}{M} n_P + S_P^+ \bar{n} + S_\perp$	$S = S_L \frac{P^-}{M} n + S^+ \bar{n} + [(S_L \frac{P_T}{M} + S_\perp) + \frac{P_T \cdot S_\perp}{P^-} \bar{n}]$
$S_\perp$		$S_\perp = S_T - S_L \frac{P_T}{M} - \frac{1}{P^-}(P_T \cdot S_T - S_L \frac{P_T^2}{M})\bar{n}$
k	$k = x P^- n_P + k_P^+ \bar{n} + k_\perp$	$k = x P^- n + k^+ \bar{n} + [(x P_T + k_\perp) + \frac{P_T \cdot k_\perp}{P^-} \bar{n}]$
$k_\perp$		$k_\perp = k_T - x P_T - \frac{1}{P^-}(P_T \cdot k_T - x P_T^2)\bar{n}$
$g'^{\mu\nu}_T$	$g_T^{\mu\nu} = g^{\mu\nu} - \bar{n}^\mu n_P^\nu - \bar{n}^\nu n_P^\mu$	$g_{TP}^{\mu\nu} = g_T^{\mu\nu} - \frac{P_T^\mu \bar{n}^\nu + \bar{n}^\mu P_T^\nu}{P^-} + \frac{P_T^2}{P^{-2}} \bar{n}^\mu \bar{n}^\nu$
$\varepsilon'^{\mu\nu}$	$\varepsilon_P^{\mu\nu} = \varepsilon^{\mu\nu\rho\sigma} \bar{n}_\rho n_{P\sigma}$	$\varepsilon_P^{\mu\nu} = \varepsilon^{\mu\nu} - \varepsilon^{\mu\nu\rho\sigma} \frac{P_{T\rho} \bar{n}_\sigma}{P^-} = \varepsilon^{\mu\nu} + \frac{1}{P^-}(\tilde{P}_T^\mu \bar{n}^\nu - \tilde{P}_T^\nu \bar{n}^\mu)$

Def $\Phi_{ab}(x, k_\perp; P, S)$

$$= \int \frac{d\xi^+ d^2\xi_T}{(2\pi)^3} e^{i(xP^-\xi^+ - k_\perp \cdot \xi_\perp)} \langle P, S | \bar{\psi}_b(0) W(0,\xi) \psi_a(\xi) | P, S \rangle \Big|_{\xi^- = 0}$$

$$= \frac{1}{2}\left\{ f_1 \not{n}_P - f_{1T}^\perp \frac{\varepsilon_{\mu\nu\rho\sigma} \gamma^\mu n_P^\nu k_\perp^\rho S_\perp^\sigma}{M} + g_{1s} \gamma_5 \not{n}_P + h_{1T} i\sigma^{n_P S_\perp} \gamma^5 + \frac{h_{1s}^\perp}{M} i\sigma^{n_P k_\perp} \gamma^5 - \frac{h_1^\perp}{M} \sigma^{k_\perp n_P} \right\}$$

$$= \frac{1}{2}\Big\{ [f_1 - f_{1T}^\perp \frac{\tilde{k}_\perp \cdot S_\perp}{M}] \not{n}_P + g_{1s} \gamma^5 \not{n}_P + [h_{1T} S_{\perp\mu} + h_{1s}^\perp \frac{k_{\perp\mu}}{M} + h_1^\perp \frac{\tilde{k}_{\perp\mu}}{M}] i\sigma^{n_P \mu} \gamma^5$$

, where

$$g_{1s} = S_L g_{1L} - \frac{k_\perp \cdot S_\perp}{M} g_{1T}, \quad h_{1s}^\perp = S_L h_{1L}^\perp - \frac{k_\perp \cdot S_\perp}{M} h_{1T}^\perp$$

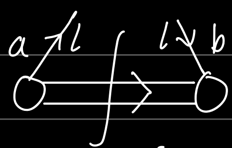
Prop

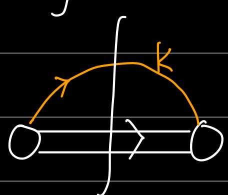
$\Phi^{\bar{\slashed{n}}}$	$\Phi^{\bar{\slashed{n}}\gamma_5}$	$\Phi^{i\sigma^{\bar{n}k}\gamma_5}$	$\Phi^{i\sigma^{\bar{n}\hat{k}}\gamma_5}$	$\Phi^{i\sigma^{\bar{n}s}\gamma_5}$
$f_1 - f_{1T}^{\perp}\dfrac{\hat{k}_\perp \cdot s_\perp}{M}$	g_{1s}	$-(h_{1T} s_\perp \cdot k_\perp + h_{1s}^{\perp}\dfrac{k_\perp^2}{M})$	$-(h_{1T} s_\perp \cdot \tilde{k}_\perp + h_1^{\perp}\dfrac{\tilde{k}_\perp^2}{M})$	$-h_{1T} s_\perp^2 - \frac{1}{M}(h_{1s}^{\perp} k_\perp \cdot s_\perp + h_1^{\perp}\hat{k}_\perp \cdot s_\perp)$

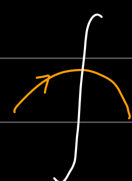
Cut - Vertex

$$\Phi = \underbrace{\int \frac{d^4 l}{(2\pi)^4}\, \delta(l^- - k^-)\,\delta^2(\vec{l}_\perp - \vec{k}_\perp)}_{\text{cut vertex}} \underbrace{\int d^4\xi\, e^{il\cdot\xi} \langle P,S|\bar{\psi}_b(0)\, W(0,\xi)\, \psi_a(\xi)|P,S\rangle}_{\hat{T}(k;P,S)}$$

Diagramatically,

$\hat{T}_{ab}(l;P,S) =$ 

$\Phi^{\Gamma}(x, k_\perp; P; S) =$ 

, where  $= \frac{1}{2}\delta(l^- - k^-)\,\delta^2(\vec{l}_\perp - \vec{k}_\perp)\,\Gamma$

$$\Phi(x, \vec{k}_\perp; P, S) = \int \frac{d^4 l}{(2\pi)^4}\, \frac{1}{P^-}\,\delta\!\left(\frac{l^-}{P^-} - x\right)\delta^2(\vec{l}_\perp - \vec{k}_\perp)\, T_{\Phi}(l;P,S)$$

Quark Collinear PDF (large P^-)

Def $\Phi_{ab}(x;P,S)$

$$= \int \frac{d\xi^+}{2\pi} e^{ixP^-\xi^+} \langle P,S|\bar\psi_b(0)W(0,\xi^+)\psi_a(\xi^+)|P,S\rangle$$

$$= \frac{1}{2}\left[f_1 \not{n}_P + S_L g_{1L}\gamma^5 \not{n}_P + h_1 i\sigma^{n_P S_\perp}\gamma^5\right]$$

Prop Let $\Phi^\Gamma(x) = \frac{1}{2}\mathrm{Tr}[\Gamma\Phi(x)]$. Then,

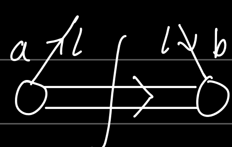
$\Phi^{\bar{\not{n}}}$	$\Phi^{\bar{\not{n}}\gamma^5}$	$\Phi^{i\sigma^{\bar{n}S}\gamma^5}$
f_1	$S_L g_{1L}$	$-H_1 S_\perp^2$

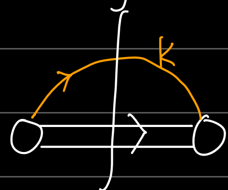
Cut - Vertex

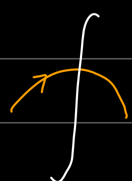
Φ_{ij}

$$= \int \frac{d^4 l}{(2\pi)^4} \underbrace{\delta(l^- - k^-)}_{\text{cut vertex}} \underbrace{\int d^4\xi\, e^{il\cdot\xi}\langle P,S|\bar\psi_b(0)W(0,\xi)\psi_a(\xi)|P,S\rangle}_{\hat{T}(k;P,S)}$$

Diagramatically,

$\hat{T}_{ab}(l;P,S) =$ 

$\Phi^\Gamma(x;P;S) =$ 

, where  $= \frac{1}{2}\delta(l^- - k^-)\Gamma$

$$\Phi(x;P,S) = \int \frac{d^4 l}{(2\pi)^4} \frac{1}{P^-}\delta\left(\frac{l^-}{P^-} - x\right) T_\Phi(l;P,S)$$

$$\int \frac{d^4 l}{(2\pi)^4} T_\Phi(l;P,S) = \int dx\, P^- \Phi(x;P,S)$$

Quark TMDFF (large k^-)

	k - Intrinsic	P - Intrinsic
n'		$n_k = n_P + \frac{k_\perp}{k^-} - \frac{k_\perp^2}{2k^{-2}}\bar{n}$
e_i'		$e_{ki} = e_{Pi} - \frac{e_{Pi}\cdot k_\perp}{k^-}\bar{n}$
k	$k = k^- n_k + k_k^+ \bar{n}$	$k = k^- n_P + k_P^+ \bar{n} + k_\perp$
S	$S = S_L \frac{z k^-}{M} n_k + S_k^+ \bar{n} + S_{\perp k}$	$S = S_L \frac{z k^-}{M} n_P + S_P^+ \bar{n} + [(S_L \frac{z k_\perp}{M} + S_{\perp k}) + \frac{k_\perp \cdot S_{\perp k}\bar{n}}{k^-}]$
$S_\perp$		$S_{\perp k} = S_\perp - S_L \frac{z k_\perp}{M} - \frac{1}{k^-}(k_\perp \cdot S_\perp - \frac{S_L}{z}\frac{k_\perp^2}{M})\bar{n}$
P	$P = z k^- n_k + P_k^+ \bar{n} + P_{\perp k}$	$P = z k^- n_P + P_P^+ \bar{n}$
$P_\perp$		$P_{\perp k} = -z k_\perp + z \frac{k_\perp^2}{k^-}\bar{n}$
$g_T'^{\mu\nu}$	$g_{Tk}^{\mu\nu} = g^{\mu\nu} - \bar{n}^\mu n_k^\nu - \bar{n}^\nu n_k^\mu$	$g_{TP}^{\mu\nu} = g_{TP}^{\mu\nu} - \frac{k_\perp^\mu \bar{n}^\nu + \bar{n}^\mu k_\perp^\nu}{k^-} + \frac{k_\perp^2}{k^{-2}}\bar{n}^\mu \bar{n}^\nu$
$\varepsilon'^{\mu\nu}$	$\varepsilon_k^{\mu\nu} = \varepsilon^{\mu\nu\rho\sigma}\bar{n}_\rho n_{k\sigma}$	$\varepsilon_P^{\mu\nu} = \varepsilon^{\mu\nu} - \varepsilon^{\mu\nu\rho\sigma}\frac{k_{\perp\rho}\bar{n}_\sigma}{k^-} = \varepsilon^{\mu\nu} + \frac{1}{k^-}(\tilde{k}_\perp^\mu \bar{n}^\nu - \tilde{k}_\perp^\nu \bar{n}^\mu)$

Note that we can define $P_\perp^\mu = g_{TP}^{\mu\nu} P_{\perp k \nu} = -z k_\perp^\mu$. Thus,

$$P_\perp = -z k_\perp$$

Definition

Consider in P-intrinsic basis $\{\bar{n}, n_P\}$

$$P = z k^- n_P + P_P^+ \bar{n}$$
$$k = k^- n_P + k_P^+ \bar{n} + k_\perp$$
$$S = \frac{S_L}{z}\frac{k}{M} n_P + S_P^+ \bar{n} + S_\perp$$

Def $\Delta_{ij}(z, P_\perp = -z k_\perp; P, S)$

$$= \frac{1}{2z} \int dPS_X \int \frac{dy^+ d^2 y_\perp}{(2\pi)^3} e^{i(k^- y^+ - \vec{k}_\perp \cdot \vec{y}_\perp)} \langle 0 | W(\infty, y) \psi_i(y) | P, S; X \rangle \langle P, S; X | \bar{\psi}_j(0) W(0, \infty) | 0 \rangle \big|_{y^- = 0}$$

$$= \frac{1}{2}\left\{ D_1 \not{n}_P - D_{1T}^\perp \frac{\varepsilon_{\mu\nu\rho\sigma}\gamma^\mu n_P^\nu P_\perp^\rho S_\perp^\sigma}{M} + G_{1s}\gamma^5 \not{n}_P + H_{1T} i\sigma^{n_P S_\perp}\gamma^5 + \frac{H_{1s}^\perp}{M} i\sigma^{n_P P_\perp}\gamma^5 - \frac{H_1^\perp}{M}\sigma^{P_\perp n_P} \right\}$$

$$= \frac{1}{2}\left\{ \left[D_1 + D_{1T}^\perp \frac{z \hat{k}_\perp \cdot S_\perp}{M}\right] \not{n}_P + G_{1s}\gamma^5 \not{n}_P + \left[H_{1T} S_{\perp\mu} - \frac{z}{M}\left(H_{1s}^\perp k_{\perp\mu} + H_1^\perp \hat{k}_\perp\right)\right] i\sigma^{n_P \mu}\gamma^5 \right.$$

, where

$$G_{1s} = S_L G_{1L} + z\frac{k_\perp \cdot S_\perp}{M} G_{1T}, \quad H_{1s}^\perp = S_L H_{1L}^\perp + z\frac{k_\perp \cdot S_\perp}{M} H_{1T}^\perp$$

Prop Let $\Delta^\Gamma(z, k_\perp) = \frac{1}{2}\mathrm{Tr}[\Gamma \Delta(z, k_\perp)]$. Then,

$\Phi^{\bar{\not{n}}}$	$\Phi^{\bar{\not{n}}\gamma^5}$	$\Phi^{i\sigma^{\bar{n}k}\gamma^5}$	$\Phi^{i\sigma^{\bar{n}\hat{k}}\gamma^5}$	$\Phi^{i\sigma^{\bar{n}S}\gamma^5}$
$D_1 + D_{1T}^\perp \frac{z\hat{k}_\perp \cdot S_\perp}{M}$	G_{1s}	$-H_{1T} S_\perp \cdot k + \frac{z}{M} H_{1s}^\perp k_\perp^2$	$-H_{1T} S_\perp \cdot \hat{k} + \frac{z}{M} H_{1s}^\perp \tilde{k}_\perp^2$	$-H_{1T} S_\perp^2 + \frac{z}{M}(H_{1s}^\perp k_\perp \cdot S_\perp + H_1^\perp \hat{k}_\perp \cdot S_\perp)$

Normalization

Prop At LO partonic level, $D_1 = G_{1L} = H_{1T} = \frac{1}{2}\delta(1-z)\delta^2(\vec{k}_\perp)$, while others are 0.
The different normalization is because we average over initial spins, but sum over final.

proof Recall $\begin{cases} \psi_i(y)|q(P,S)\rangle = u_i(P,S)e^{-iP\cdot y}|0\rangle \\ \langle q(P,S)|\bar{\psi}_j(y) = \bar{u}_j(P,S)e^{iP\cdot y}\langle 0| \end{cases}$

$\Delta_{ij}(x, k_\perp; P, S)$

$= \frac{1}{2z}\int dPS_X \int \frac{dy^+ d^2y_\perp}{(2\pi)^3} e^{i(k^- y^+ - \vec{k}_\perp\cdot\vec{y}_\perp)} \langle 0|\psi_i(y)|P,S;X\rangle\langle P,S;X|\bar{\psi}_j(0)|0\rangle|_{y^-=0}$

$= \frac{1}{2z}\int \frac{dy^+ d^2y_\perp}{(2\pi)^3} e^{i(k^- y^+ - \vec{k}_\perp\cdot\vec{y}_\perp)}[u_S(P)\bar{u}_S(P)]_{ij} e^{-ip^- y^+} = \frac{1}{2z}[u_S(k)\bar{u}_S(k)]_{ij}\,\delta(k^- - P^-)\,\delta(\vec{k}_\perp)$

$= \frac{1}{4P^-}(\not{P} + m)(1 + \gamma^5 \not{S})\,\delta(1-z)\delta^2(\vec{k}_\perp)$

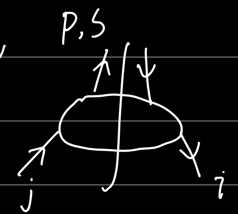
This is exactly half of the TMDPDF case $\square$

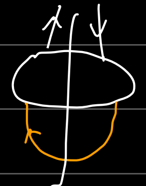
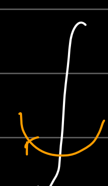
Cut - Vertex

Δ_{ij}

$= \int \frac{d^4 l}{(2\pi)^4} \underbrace{\frac{1}{2z}\delta(l^- - k^-)\delta^2(\vec{l}_\perp - \vec{k}_\perp)}_{\text{cut vertex}} \underbrace{\int dPS_X \int d^4y\, e^{il\cdot y}\langle 0|W(\infty,y)\psi_i(y)|P,S;X\rangle\langle P,S;X|\bar{\psi}_j(0)W(0,\infty)|0\rangle}_{\hat{T}_{ij}(l;P,S)}$

Diagramatically,

$\hat{T}_{ab}(l;P,S) =$ 

$\Phi^\Gamma(x, k_\perp; P; S) =$ , where  $= \frac{1}{4z}\delta(l^- - k^-)\delta^2(\vec{l}_\perp - \vec{k}_\perp)\Gamma$

$\Delta(z, k_T; P_h, S_h) = \int \frac{d^4 l}{(2\pi)^4} \frac{1}{2z}\delta\!\left(l^- - \frac{P_h^-}{z}\right)\delta^2(\vec{l}_\perp - \vec{k}_\perp)\,T_\Delta(l;P,S)$

Quark Collinear FF (large k^-)

Def $\Delta_{ij}(z;P,S)$

$$= \frac{z}{2}\int dPS_X \int \frac{dy^+}{2\pi} e^{ik^- y^+} \langle 0| W(\infty, y^+) \psi_i(y^+) |P,S;X\rangle\langle P,S;X| \bar{\psi}_j(0) W(0,\infty)|0\rangle$$

$$= \frac{1}{2}\left[D_1 \not{n}_P + S_L G_{1L} \gamma^5 \not{n}_P + H_1 i\sigma^{n_P S_\perp}\gamma^5 \right]$$

Prop Let $\Delta^\Gamma(z) = \frac{1}{2}\mathrm{Tr}[\Gamma \Delta(z)]$. Then,

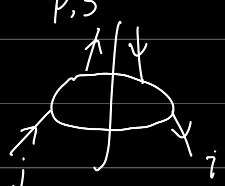
$\Phi^{\bar{\not{n}}}$	$\Phi^{\bar{\not{n}}\gamma^5}$	$\Phi^{i\sigma^{\bar{n}S}\gamma^5}$
D_1	$S_L G_{1L}$	$-H_1 S_\perp^2$

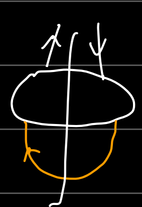
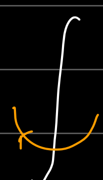
Cut-Vertex

Δ_{ij}

$$= \int \frac{d^4 l}{(2\pi)^4} \underbrace{\frac{z}{2}\delta(l^- - k^-)}_{\text{cut vertex}} \underbrace{\int dPS_X \int d^4 y\, e^{il\cdot y} \langle 0| W(\infty,y)\psi_i(y)|P,S;X\rangle\langle P,S;X|\bar{\psi}_j(0) W(0,\infty)|0\rangle}_{\hat{T}_{ij}(l;P,S)}$$

Diagramatically,

$\hat{T}_{ab}(l;P,S) =$ 

$\Delta^\Gamma(x, k_\perp; P; S) =$ , where  $= \frac{z}{4}\delta(l^- - k^-)\Gamma$

$$\Delta_{ij}(z;P,S) = \int \frac{d^4 l}{(2\pi)^4}\frac{z}{2}\delta\left(l^- - \frac{P^-}{z}\right)\hat{T}_{ij}(l;P,S) = \int \frac{d^4 l}{(2\pi)^4}\frac{z^3}{2P^-}\delta\left(z - \frac{P^-}{l^-}\right)\hat{T}_{ij}(l;P,S)$$

$$\int \frac{d^4 l}{(2\pi)^4}\hat{T}_{ij}(l;P,S) = \int dz\, \frac{2P^-}{z^3}\Delta_{ij}(z;P,S)$$